Volume Fusion-Based Self-Supervised Pretraining for 3D Medical Image Segmentation

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***Abstract*—The performance of deep learning models for medical image segmentation is often limited in scenarios where training data or annotations are limited. Self-Supervised Learning (SSL) is an appealing solution for this dilemma due to its feature learning ability from a large amount of unannotated images. Existing SSL methods have focused on pretraining either an encoder for global feature representation or an encoder-decoder structure for image restoration, where the gap between pretext and downstream tasks limits the usefulness of pretrained decoders in downstream segmentation. In this work, we propose a novel SSL strategy named Volume Fusion (VolF) for pretraining 3D segmentation models. It minimizes the gap between pretext and downstream tasks by introducing a pseudo-segmentation pretext task, where two sub-volumes are fused by a discretized block-wise fusion coefficient map. The model takes the fused result as input and predicts the category of fusion coefficient for each voxel, which can be trained with standard supervised segmentation loss functions without manual annotations. Experiments with an abdominal CT dataset for pretraining and both in-domain and out-domain downstream datasets showed that VolF led to large performance gain from training from scratch with faster convergence speed, and outperformed several state-of-the-art SSL methods. In addition, it is general to different network structures, and the learned features have high generalizability to different body parts and modalities.**

***Index Terms*—Self-supervised learning, feature representation, segmentation, computed tomography.**

I. Introduction

DEEP learning models such as Convolutional Neural Networks (CNNs) and Transformers have achieved state-of-the-art performance in automatic medical image segmentation that is essential for computer-assisted diagnosis and treatment planning [1]. Their high performance heavily relies on supervised learning from a large set of annotated training images [2], [3]. However, annotating 3D medical images with pixel-level labels requires expert knowledge and is time-consuming, which makes it expensive and difficult to obtain segmentation annotations for a large training set. In addition, in many clinical scenarios, the available training set is small, making it hard to train a high-performance segmentation model [4]. To deal with these problems, annotation-efficient learning methods have been widely studied, such as weakly-supervised learning, semi-supervised learning [5], [6], domain adaptation [7] and noisy-label learning [8], etc. Among them, Self-Supervised Learning (SSL) is an appealing method as it can learn useful features from unannotated images via pretext tasks, which helps to reduce the annotation cost and improve performance in downstream segmentation tasks after fine-tuning [9], [10].

Many existing SSL methods are based on image/patch-level classification or contrastive learning [11], [12], [13] to pretrain a feature extractor (i.e., encoder), while segmentation models usually have an additional decoder that is not used in the pretraining process of these SSL methods, which makes them less effective for segmentation. To alleviate this problem, image restoration-based pretext tasks [9], [14] pretrain an additional decoder for pixel-level intensity prediction. However, as such pretext tasks tend to over-fit low-level intensity features, they are hardly transferable to semantic segmentation tasks that rely more on global contextual features, making the pretrained decoder useless for segmentation tasks. Fig. 1(a) shows this phenomenon for a UNet-like segmentation model pretrained by Model Genesis [9] using image restoration. It can be observed that in the downstream segmentation task, loading the pretrained encoder for initialization performed better than training from scratch, but additionally loading the pretrained decoder for initialization does not bring extra performance improvement. Thus, more effective pretext tasks are needed to lean segmentation-friendly features in the decoder.

To address these issues, we propose a novel self-supervised pretraining strategy named as Volume Fusion (VolF) that is specifically designed for 3D medical image segmentation. As shown in Fig. 2, unlike existing SSL methods that define pretext tasks as image-level classification or image restoration, our method uses a simple yet effective pseudo-segmentation pretext task that has a minimal gap to downstream

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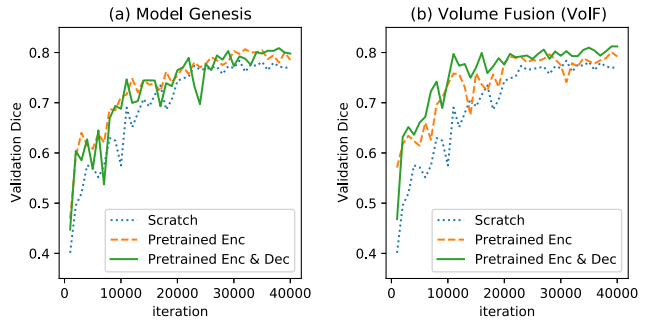

Fig. 1. Effectiveness of pretrained encoder and decoder in different SSL methods. The AbdomenCT-1K and LiTS datasets were used for self-supervised pretraining and downstream segmentation, respectively. Orange curve means only loading the pretrained encoder, while green curve means loading both the pretrained encoder and decoder for initialization in the downstream task, respectively. (a) For Model Genesis [9], the pretrained decoder does not bring extra benefits compared to only using the pretrained encoder. (b) For our VolF, both the pretrained encoder and decoder contribute to performance improvement.

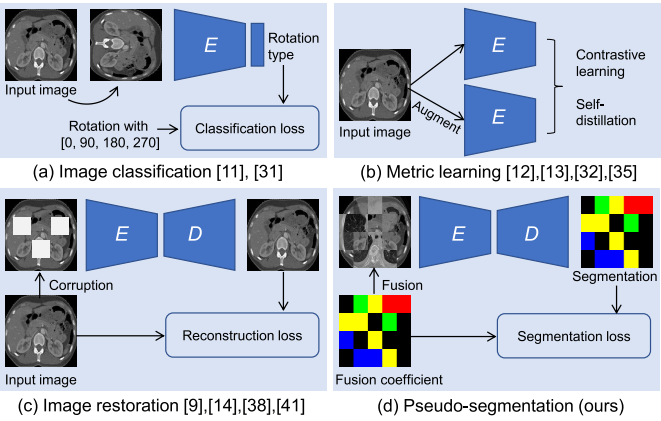

Fig. 2. Illustration of the difference between existing Self-Supervised Learning (SSL) methods and ours.

segmentation tasks. Specifically, it firstly obtains two sub-volumes from the same 3D image with intensity and spatial augmentations, and then fuses them based on a discretized block-wise fusion coefficient map. The coefficient of each block takes values from a predefined set where each element is treated as a class. The model to pretrain takes the fused sub-volume as input, and predicts the class label of each voxel, which is exactly a supervised segmentation pretext task without manual annotations. This pseudo-segmentation pretext task makes features in both encoder and decoder learned during pretraining be more transferable to downstream segmentation tasks, as shown in Fig. 1(b).

The contribution of this work is three-fold:

- First, we propose Volume Fusion (VolF), a novel SSL method to pretrain 3D medical image segmentation models. It enables pretraining both encoder and decoder of a segmentation model from unannotated images by using a pseudo-segmentation pretext task that matches well with the downstream task of segmentation.
- Second, we propose a block-wise fusion coefficient map generation method for fusion of two sub-volumes, where the fusion coefficient of each block takes values from a predefined set, and the block size is set randomly to simulate objects at different scales.
- Finally, we propose to pretrain the segmentation model with the pair of fused volume and the discretized block-wise fusion coefficient map, where standard segmentation loss can be employed for pretraining, which makes the pretraining task quite simple and compatible with downstream segmentation tasks.

In the experiments, we used 1,112 unannotated abdominal Computed Tomography (CT) volumes for pretraining and five in-domain and out-domain datasets for downstream segmentation. The results showed that 1) the proposed VolF is superior to six state-of-the-art SSL methods in terms of segmentation accuracy and convergence speed on downstream datasets; 2) both the encoder and decoder features learned by VolF contribute to performance improvement in downstream segmentation; 3) VolF is generalizable to different network structures including CNNs and Transformers, and the features learned by VolF from abdominal CT scans generalize well to images of other body parts or modalities such as chest CT and abdominal Magnetic Resonance Images (MRI).

II. Related Works

A. Medical Image Segmentation Models

Deep learning models for medical image segmentation can be roughly divided into Convolutional Neural Network (CNN)-based, Transformer-based and hybrid models that combine CNN and Transformer. CNN-based models are dominated by U-Net-like neural networks [3], [15] with new modules such as residual connection [16], attention [17], [18] and multi-scale feature learning [19]. Recently, Vision Transformers (ViT) [20], [21] based on self-attention have been increasingly investigated. For example, He et al. [22] proposed a Fully Transformer Network (FTN) to learn long-range contextual information. Swin-UNet [23] is a U-Net-like pure Transformer that uses swin Transformer based on shifted windows for improving the efficiency. UNETR++ [24] introduces parallel attention modules with shared key-queries to efficiently learn enriched spatial-channel feature representations. Some works like TransUNet [25], UTNet [26] and nnFormer [27] have been proposed to combine the advantage of CNNs and Transformers for medical image segmentation. In addition, Mamba-based models like Swin-UMamba [28] and U-Mamba [29] have been introduced for medical image segmentation with better ability to model long-range global information.

B. Self-Supervised Learning

Self-supervised learning has been widely used for visual representation learning via pretext tasks without human annotations [30]. Pretext tasks can be defined in various formulations. The first type is classification, such as prediction of discrete rotation angles [11] and jigsaw puzzle solving [31]. The second type is contrastive learning [32], [33], [34] that maximizes the similarity between positive pairs and minimizes that between negative pairs based on a contrastive loss, and it is extended to self-distillation [12], [35]. Lu et al. [36] used a

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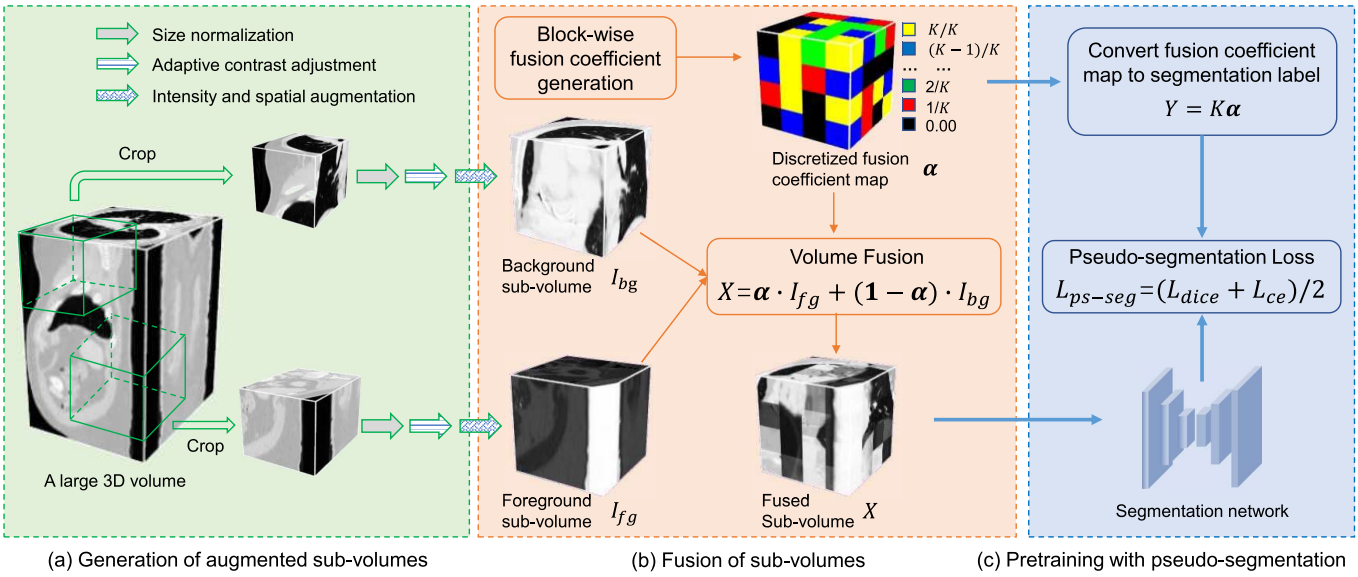

Fig. 3. Overview of our proposed self-supervised pretraining method Volume Fusion (VolF) for medical image segmentation models.

self-supervised feature comparison loss for anomaly detection, and Sun et al. [37] applied contrastive learning to atlas-based brain image segmentation, but it requires annotations of atlases for pretraining. In addition, some image restoration methods have been proposed, such as image denoising [38], [39], [40] and Masked Auto-Encoder (MAE) [14], [41].

For 3D medical images, Vox2vec [13] is a contrastive method for voxel-level representation learning, and VoCo [42] leverages contextual position priors for volume contrastive learning. Rubik's Cube+ [43] uses cube ordering, cube rotating and cube masking as pretext tasks. For pretext tasks based on image restoration, Chen et al. [44] proposed random swap of local patches to train CNNs, and Model Genesis [9] combines non-linear transformation, local pixel shuffling and outer/inner cutouts. Masked Image Modeling (MIM) [14] is a 3D extension of MAE for training vision Transformers. Tang et al. [10] combined inpainting, contrastive learning and rotation prediction to pretrain a 3D Swin Transformer. However, these works are faced with the gap between pretext task of image restoration and downstream task of segmentation, as the features optimized for low-level intensity prediction are hardly transferable to semantic segmentation tasks, especially in decoder of the model.

III. Methods

Fig. 3 shows an overview of our Volume Fusion (VolF) framework for self-supervised pretraining of 3D medical image segmentation models. It introduces a pretext task based on pseudo-segmentation to match the downstream task of segmentation. First, two sub-volumes are extracted from a 3D volume, and resized to the same size followed by intensity and spatial augmentation. Then, a discretized block-wise fusion coefficient map is generated to fuse the two sub-volumes, where the fusion coefficient value of each block is taken from a predefined set. Finally, the segmentation model is pretrained in a standard supervised learning paradigm, where the fused sub-volume serves as the input, and the fusion coefficient map is converted into a pseudo-segmentation label map for supervision.

A. Sub-Volume Generation and Augmentation

As shown in Fig. 3(a), for 3D volumes in the unlabeled pretraining dataset, we obtain two sub-volumes for fusion. Note that the two sub-volumes may be from two different volumes in the training set, and for implementation simplicity in this work, we obtain the two sub-volumes at different positions in a single volume with random cropping. Let D, H and W denote the depth, height and width of the target shape of the fused sub-volume, respectively. The crop size along each dimension is a random number in the range of $(1/a, a)$ times of the target size, and $a \geq 1.0$. The introduction of a helps to cropping the volume at different scales, leading to scale augmentation, and we set $a = 2.0$ in this work. The initial two sub-volumes are denoted as I_0 and I_1, respectively.

Then, I_0 and I_1 are resized to the same shape of $D \times H \times W$. To improve the generalizability of learned features during pretraining, we apply several spatial and intensity augmentations before fusion of the two sub-volumes. In addition to the above resize-based cropping, we also use random flipping and transpose for spatial augmentation, and random contrast adjustment, gamma correction and intensity inversion for intensity augmentation. Specifically, for random contrast adjustment, let i_{min} and i_{max} denote the minimal and maximal intensity value of a sub-volume, and i_p and i_q denote the p-th and q-th percentiles ($p < q$), respectively, e.g., $p = 1$ and $q = 99$. As illustrated in Fig. 4(c), we use a segmented linear function φ to map the range (i_{min}, i_p) to (0.0, 0.1), (i_p, i_q) to (0.1, 0.9) and (i_q, i_{max}) to (0.9, 1.0), respectively. Fig. 4(d) shows that using φ for normalization leads to better contrast than min-max normalization, which better matches the pre-processing in downstream segmentation tasks where the contrast is usually adjusted for target tissues. By setting

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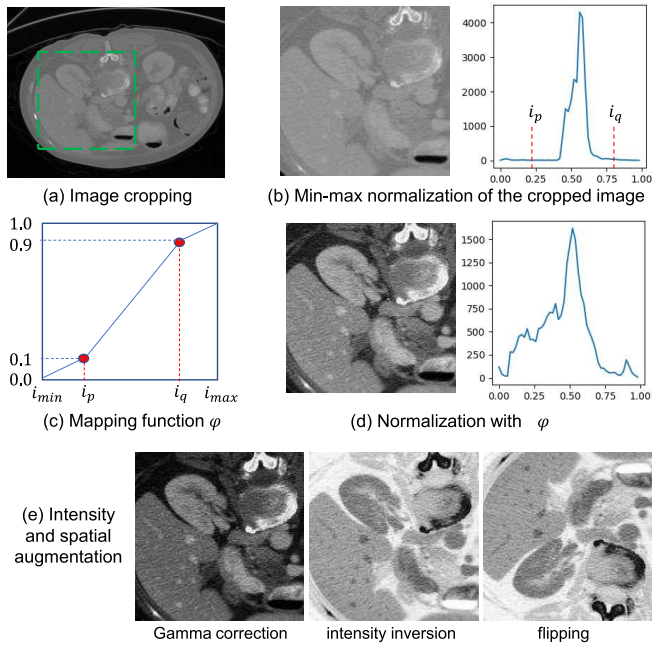

Fig. 4. A 2D illustration of sub-volume generation and augmentation.

p and q randomly, different contrasts can be obtained for augmentation.

B. Fusion of Sub-Volumes

The volume fusion process is formulated as replacing a sub-region of a background image I_{bg} by a mixture of that region and the corresponding region in another image I_{fg}, i.e., foreground image. For simplicity, we use the resized and augmented versions of I_0 and I_1 as I_{fg} and I_{bg}, respectively. Note that I_0 and I_1 are not ordered during their generation, and their role can be switched during the fusion process.

Given I_{bg} and I_{fg} with the same shape of $D \times H \times W$, we use $\boldsymbol{\alpha} \in [0,1]^{D\times H\times W}$ to denote the fusion coefficient map, and the fused output is formulated as:

$$X = \boldsymbol{\alpha} I_{fg} + (\mathbf{1} - \boldsymbol{\alpha}) I_{bg} \tag{1}$$

In the pretraining process, we send X to a segmentation model to predict $\boldsymbol{\alpha}$. To model the pretext task as a pseudo-segmentation task, the value of each voxel in $\boldsymbol{\alpha}$ is taken from a predefined set $\mathcal{V} = \{0.0, 1/K, 2/K, \ldots, (K-1)/K, 1.0\}$, where hyper-parameter K is the number of non-zero fusion coefficient values, and it corresponds to the foreground class number in our pseudo-segmentation pretext task. Note that $\boldsymbol{\alpha}$ is randomly generated for each pair of I_{fg} and I_{bg}, as detailed in the following.

C. Block-Wise Fusion Coefficient

Considering that the label of neighboring pixels are not independent in segmentation tasks, we use a block-wise fusion coefficient generation strategy to obtain $\boldsymbol{\alpha}$. Specifically, the coordinate space of $\boldsymbol{\alpha}$ is split into non-overlapping blocks with a block size of $d \times h \times w$. Note that the block size is randomly set in a large range to simulate segmentation targets at multiple scales in the pseudo-segmentation pretext task, which enables the model to effectively learn multi-scale features for dealing with segmentation targets at various scales in downstream segmentation datasets.

With a random block size, we use $\boldsymbol{\alpha}_b$ to denote the fusion coefficient for the b-th block, and assign it with a random value from the predefined set $\mathcal{V}$. Note that $\boldsymbol{\alpha}_b = 0$ means that the b-th block of I_{bg} is not affected by I_{fg} during fusion, so we call the blocks in X with $\boldsymbol{\alpha}_b = 0$ as background region, and those with non-zero $\boldsymbol{\alpha}_b$ values as foreground region.

To control the size of foreground region in X, we introduce another hyper-parameter $r \in (0,1)$ to denote the foreground ratio. Specifically, $\boldsymbol{\alpha}_b$ is set to 0 with a probability of 1-r, otherwise it is randomly taken from $\{1/K, 2/K, \ldots, (K-1)/K, 1.0\}$ with an equal probability for each of the K values. An example of the discretized block-wise fusion coefficient map $\boldsymbol{\alpha}$ is shown in Fig. 3(b), and a 2D illustration of X and $\boldsymbol{\alpha}$ with different K, r and block sizes is shown in Fig. 5.

D. Pretext Task Based on Pseudo-Segmentation

With the discretized block-wise fusion coefficient map $\boldsymbol{\alpha}$ and the fused sub-volume X, as the possible values in $\boldsymbol{\alpha}$ is in the predefined set of $\mathcal{V}$, we take each element in $\mathcal{V}$ as a class, leading to $C = K + 1$ classes of fused voxels in X. We use Y to denote the class label map corresponding to $\boldsymbol{\alpha}$, and the i-th voxel's class label y_i is 0, 1, 2, …, K for α_i = 0.0, $1/K$, $2/K$, …, 1.0, respectively. Therefore, the discretized fusion coefficient map is converted to a segmentation label map by:

$$Y = K\boldsymbol{\alpha} \tag{2}$$

With the pair of X and Y, a pseudo-segmentation loss L_{ps-seg} based on Dice loss L_{dice} and cross entropy loss L_{ce} is used to pretrain the segmentation model:

$$L_{ps-seg} = \frac{1}{2}(L_{dice} + L_{ce}) \tag{3}$$

$$L_{dice} = 1 - \frac{1}{C}\sum_{c=0}^{C-1} \frac{\sum_{i=0}^{V-1} 2 p_{i,c} y_{i,c}}{\sum_{i=0}^{V-1}(p_{i,c} + y_{i,c} + \epsilon)} \tag{4}$$

$$L_{ce} = -\frac{1}{V}\sum_{i=0}^{V-1}\sum_{c=0}^{C-1} y_{i,c} log(p_{i,c}) \tag{5}$$

where $V = DHW$ is the voxel number, and $\epsilon = 10^{-5}$ is a small number for numeric stability. $p_{i,c}$ is the predicted probability of voxel i being class c, and $y_{i,c}$ is the corresponding one-hot ground truth value obtained from Y. The pseudo-code of our method is shown in Algorithm 1.

E. Context Learning Ability of VolF

As the ability to recognize contextual semantics is essential for medical image segmentation [9], [44], a good pretraining strategy for segmentation models should have high context learning ability. In VolF, the model is trained to recognize the fusion category of each voxel. For a given fused voxel, as there are numerous possible combinations of the original foreground and background intensities, one cannot directly find a mapping from a fused voxel's intensity to the fusion

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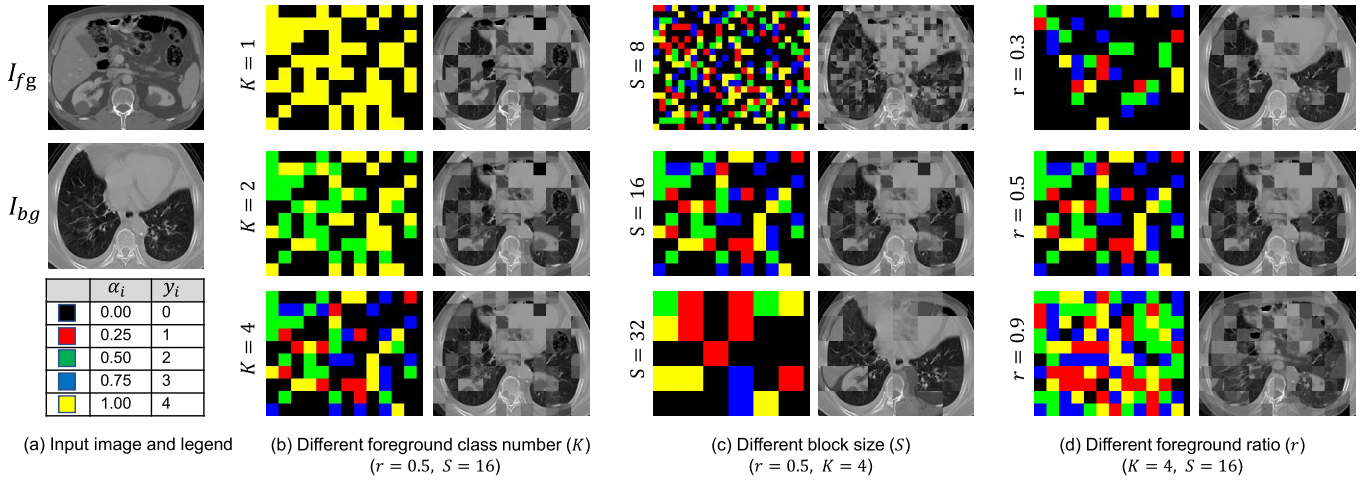

(a) Input image and legend (b) Different foreground class number (K) ($r = 0.5,\ S = 16$) (c) Different block size (S) ($r = 0.5,\ K = 4$) (d) Different foreground ratio (r) ($K = 4,\ S = 16$)

Fig. 5. 2D illustration of fused images with different foreground class number (K), block size (S) and foreground ratio (r). In (b), (c) and (d), the two columns show fusion coefficient map $\boldsymbol{\alpha}$ and output X, respectively. α_i is the fusion coefficient of pixel i, and y_i is the class index.

Algorithm 1 SSL Based on VolF

Input: Unannotated dataset $\mathcal{D}$, learning rate η, batch size N, foreground ratio r, foreground class number K.

Output: Pretrained network parameter θ.

```
Randomly initialize θ
while Stop condition not reached do
    Initialize training batch B = ∅
    for n = 0, 1, ..., N − 1 do
        I ← A 3D volume in D
        I_0, I_1 ← Obtain two sub-volumes from I
        I_bg ← Resize and data augmentation of I_0
        I_fg ← Resize and data augmentation of I_1
        Initialize α as zero array
        B ← Get block number according to random
          block size
        for block index b = 0, 1, ..., B − 1 do
            if random() < r then
                α_b = randint(1, K)/K
            end if
        end for
        X ← α I_fg + (1 − α) I_bg according to Eq. (1)
        Y ← K α according to Eq. (2)
        B ← B ∪ {(X, Y)}
    end for
    Forward pass with the batch B
    L_ps−seg ← Pseudo-segmentation loss via Eq. (3)
    θ ← θ − η ∇_θ L_ps−seg
end while
```

category. Successful recognition of the fusion category of a voxel requires being aware of the semantic information of the foreground and background sub-volumes.

Fig. 5(b) shows an example of fused images with different K values. When $K = 1$, the model is trained for binary segmentation of the foreground blocks from the background region. As both of the foreground and background sub-volumes are CT scans with similar intensity distributions, the model needs to find the structural discontinuity around the border of blocks for the segmentation, which forces it to have a context learning ability. When $K > 1$, there are more types of foreground blocks with soft fusion coefficients, and the model needs a higher contextual semantics recognition ability for the segmentation task. As shown in Fig. 5(b), higher K values lead to more classes in the pseudo-segmentation pretext task, which increases difficulty for pretraining and helps to improve the feature learning ability.

Fig. 5(c) and Fig. 5(d) show the fused image under different block sizes and foreground ratios, respectively. It can be observed that a small r value leads to a few 'outlier' (foreground) blocks in the image that are relatively easy to recognize. Using a relatively larger r value leads to higher difficulty for the pretext task, which helps to improve the feature representation ability of the pretrained model.

IV. Experiments and Results

A. Datasets for Pretraining and Downstream Tasks

As listed in Table I, we used the **AbdomenCT-1K dataset** [45] containing 1,112 abdominal CT scans for self-supervised pretraining. For downstream segmentation tasks, we considered five datasets for organ and lesion segmentation from CT and MRI.

1) In-Domain Downstream Datasets: We first used three small abdominal CT datasets that are similar to the pretraining dataset for downstream segmentation: **1) Liver Tumor Segmentation (LiTS)** [46] challenge 2017 dataset for liver and liver tumor segmentation, containing publicly available and annotated CT scans from 131 patients with liver cancer. The data was randomly split into 6:2:2 for training, validation and testing. **2) Kidney Tumor Segmentation (KiTS)** [47] dataset for kidney and kidney tumor segmentation. It provides 210 publicly available CT scans with annotations. They were randomly split into 120, 30 and 60 for training, validation and testing, respectively. **3) Synapse dataset** that includes 30 abdominal CT scans for organ segmentation. Following

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TABLE I

DATASETS USED FOR EXPERIMENTS, WHERE 3 IN-DOMAIN AND 2 OUT-DOMAIN DOWNSTREAM DATASETS ARE USED, RESPECTIVELY. THE △ SHOWS UNSEEN MODALITY OR STRUCTURE DURING PRETRAINING

Stage	Pretrain	Downstream (In-domain)			Downstream (Out-domain)	
Dataset	AbdomenCT-1K	LiTS	KiTS	Synapse	SegTHOR	AMOS-MRI
Modality	CT	CT	CT	CT	CT	MRI (△)
Body part	Abdomen	Abdomen	Abdomen	Abdomen	Chest (△)	Abdomen
Size (volumes)	1,112	131	282	30	40	110
Target	None	Liver tumor	Kidney tumor	Organs	Organs	Organs

existing works [25], [27], we deal with 8 organs, i.e., aorta, gallbladder, left kidney, right kidney, liver, pancreas, spleen and stomach. The 3D images were randomly split into 18, 4 and 8 for training, validation and testing, respectively.

2) Out-Domain Downstream Datasets: To investigate the transferability of the features learned by SSL to unseen body parts or modalities, we considered two out-domain downstream datasets where the body part or modality is different from the pretraining set: **1) SegTHOR dataset** [48] with 60 thoracic CT scans for segmentation of thoracic Organs-at-Risk (OARs), including the heart, aorta, trachea and esophagus. As annotations for the official testing set (20 cases) are not publicly available, we used the remaining 40 cases for experiments, with 20, 5 and 15 for training, validation and testing, respectively. **2) Abdominal MRI segmentation dataset** that is a combination of the AMOS-MRI dataset [49] (60 MRI volumes) and another set of 50 MRI volumes provided by Ma et al. [29]. Following [29], we used 40 and 20 cases from the AMOS-MRI dataset for training and validation, and the extra 50 volumes from Ma et al. [29] for testing, respectively, and segmented 13 organs including the liver, right kidney, spleen, pancreas, aorta, Inferior Vena Cava (IVC), Right Adrenal Gland (RAG), Left Adrenal Gland (LAG), gallbladder, esophagus, stomach, duodenum and left kidney.

B. Implementation Details

The code for pretraining based on VolF is available in PyMIC[1] [50] based on PyTorch. Tutorials for using VolF and the pretrained models are available online.[2] Our pre-trained models are also deployed on the SenseCare [51] platform for clinical research. In this work, the experiments were conducted on a Ubuntu desktop with an NVIDIA GTX 3090 GPU. The segmentation network was a modification of 3D U-Net [52], where we used instance normalization and Leaky ReLU in convolutional layers, and strided convolution for down-sampling. The network has five resolution levels, with feature channel number of 32, 64, 128, 256 and 512, respectively. The modified network is named as FM-UNet, and the model size is 47.37 M. For pretraining, the sub-volume size was $64 \times 128 \times 128$. To generate random fusion coefficient map $\boldsymbol{\alpha}$, the foreground class number was $K = 4$, and the block size was a random number in the range of $4 \times 8 \times 8$ to $16 \times 32 \times 32$. The foreground ratio was $r = 0.7$. For pretraining, we used a batch size of 2 with the Adam optimizer, weight decay of 10^{-5} and total iteration number of 100k. The learning rate was initialized to 10^{-3} and reduced by half every 20k iterations.

[1]https://github.com/HiLab-git/PyMIC/tree/master/pymic/net_run/self_sup

[2]https://github.com/openmedlab/MIS-FM

The combination of Dice loss and cross entropy loss was used for both the pretraining and downstream tasks. In each downstream task, we loaded both the pretrained encoder and decoder for initialization, and tuned all the layers. The total training iteration was 20k for SegTHOR and Synapse datasets, and 40k for the other three datasets. Random resizing, cropping, flipping and gamma correction were used for data augmentation. A batch size of 2 was used, and the learning rate was initialized to 10^{-3} with a polynomial learning rate scheduler and the Adam optimizer. For quantitative evaluation, we used the Dice similarity coefficient (Dice) and Average Symmetric Surface Distance (ASSD) between segmentation results and the corresponding ground truths.

C. Comparison With State-of-the-Art SSL Methods

Our proposed VolF was compared with six existing SSL methods for medical image segmentation: 1) Patch Swap [44], 2) Model Genesis [9], 3) Masked Image Modeling (MIM) [14], 4) DINO [12] that is a self-distillation method for pretraining with a mean teacher structure, 5) VoCo [42] that is a volume contrast framework using contextual position priors, and 6) Vox2vec [13] based on voxel-wise contrastive learning. All these methods were implemented with the same backbone network of our FM-UNet. Note that the three image restoration-based methods (Patch Swap, Model Genesis and MIM), we only loaded the pretrained encoder for initialization, as additionally loading the pretrained decoder for initialization does not bring performance gain and sometimes even leads to poorer performance, according to Fig. 1.

1) Results on In-Domain Downstream Datasets: We first compared the different SSL methods on the LiTS, KiTS and Synapse datasets. The results for LiTS and KiTS are shown in Table II. On the LiTS dataset, training from scratch obtained an average Dice of 65.42% for liver tumor. All the SSL methods improved the performance to some degree, and MIM [14] slightly outperformed Patch Swap [44] and Model Genesis [9] among the restoration-based SSL methods. DINO [12] and VoCo [42] performed inferior to the other methods, while Vox2vec [13] performed the best among the existing SSL methods, with a Dice of 71.34% for liver tumor. Compared with training from scratch, our VolF improved the average Dice by more than 7 percentage points from 65.42% to 72.60%, and it also outperformed Vox2vec [13]. The ASSD value of liver tumor obtained by VolF was 3.98 mm, which was much better than training from scratch (6.23 mm) and the other SSL methods. From the evolution of validation Dice on the LiTS dataset in Fig. 1, it can be observed that VolF does not only led to better segmentation performance after

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TABLE II

QUANTITATIVE EVALUATION OF SSL METHODS ON LITS AND KITS DATASETS. THE BEST AND SECOND-BEST RESULTS IN EACH COLUMN ARE IN BOLD AND UNDERLINED, RESPECTIVELY

Method	Dice (%)				ASSD (mm)			
	Liver	Liver tumor	Kidney	Kidney tumor	Liver	Liver tumor	Kidney	Kidney tumor
Scratch	94.91±2.83	65.42±22.05	92.25±7.76	70.72±18.81	1.51±0.53	6.23±7.49	2.67±3.77	9.08±7.75
Patch Swapping [44]	95.00±2.50	68.03±17.83	92.88±8.36	74.22±21.30	1.68±1.10	5.93±6.29	2.51±3.96	6.57±6.85
Model Genesis [9]	95.10±2.71	69.32±18.16	92.90±7.93	73.09±21.87	1.53±0.54	5.62±6.11	2.37±3.37	6.10±6.57
MIM [14]	95.00±2.71	69.73±15.00	93.01±7.19	76.14±14.04	1.52±0.58	5.67±6.97	2.48±3.76	6.52±7.26
DINO [12]	94.78±2.98	66.92±18.06	92.92±7.09	73.49±23.12	1.56±0.57	6.17±5.64	2.56±4.05	6.71±7.14
VoCo [42]	95.25±2.74	67.39±21.41	93.18±5.97	75.80±18.29	1.36±0.47	5.65±5.61	2.15±2.82	6.02±6.64
Vox2vec [13]	95.03±2.93	71.34±14.29	93.51±6.16	77.72±16.70	1.58±0.46	4.15±4.62	1.97±2.96	5.99±6.98
VolF (Ours)	**95.49±2.84**	**72.60±15.56**	**93.75±9.23**	**79.31±16.91**	**1.24±0.41**	**3.98±3.93**	**1.42±2.03**	**4.60±5.79**

TABLE III

QUANTITATIVE COMPARISON BETWEEN DIFFERENT SSL METHODS FOR ABDOMINAL ORGAN SEGMENTATION ON SYNAPSE DATASET. 'R' AND 'L' REPRESENT RIGHT AND LEFT, RESPECTIVELY. DICE (%) AND ASSD (MM) VALUES ARE LISTED IN THE TWO SECTIONS, RESPECTIVELY. THE BEST AND SECOND-BEST RESULTS IN EACH COLUMN ARE IN BOLD AND UNDERLINED, RESPECTIVELY

Method	Spleen	R Kidney	L Kidney	Gallbladder	Pancreas	Liver	Stomach	Aorta	Average
Scratch	93.91±2.16	92.92±2.83	91.38±5.67	68.49±26.22	72.35±12.14	95.24±2.39	87.49±3.32	87.25±5.86	86.13±5.31
Patch Swap [44]	95.43±0.82	**94.10±1.62**	92.97±3.81	66.10±28.76	78.39±6.23	95.74±1.62	87.23±5.12	90.08±4.11	87.51±4.78
Model Genesis [9]	93.44±3.53	93.21±2.86	92.13±5.53	68.95±26.69	76.21±7.17	94.96±3.76	86.39±6.54	89.04±4.83	86.79±4.54
MIM [14]	95.30±0.96	93.88±1.77	92.27±4.65	68.57±27.03	77.28±7.10	95.32±2.28	88.37±2.83	89.01±6.07	87.50±4.83
DINO [12]	94.43±1.65	94.08±1.42	92.18±4.77	67.72±28.16	77.99±5.64	95.63±1.33	87.52±3.99	89.06±4.73	87.32±4.73
VoCo [42]	95.22±1.29	93.56±2.38	92.34±5.12	71.42±26.95	76.09±7.26	95.51±1.84	89.00±3.47	89.88±4.47	87.88±4.77
Vox2Vec [13]	94.95±1.11	93.96±1.72	92.39±5.09	**72.24±26.64**	76.04±7.84	95.58±1.72	89.16±2.87	**90.14±4.50**	88.06±4.57
VolF (Ours)	**95.49±1.08**	93.39±2.75	**93.00±4.57**	70.99±27.16	**80.40±4.94**	**95.87±2.11**	**90.59±2.55**	89.66±4.51	**88.67±4.37**
Scratch	2.41±3.14	0.83±0.32	1.10±0.76	8.32±10.91	3.18±1.94	4.49±9.24	4.71±5.19	1.82±1.80	3.36±3.29
Patch Swap [44]	2.23±3.10	0.80±0.28	0.97±0.70	6.12±7.11	2.31±0.97	3.17±5.70	5.43±7.36	1.28±0.55	2.79±2.37
Model Genesis [9]	1.88±2.86	0.96±0.46	1.02±0.73	5.48±5.54	3.41±2.81	3.42±4.32	5.04±5.94	1.35±1.03	2.82±2.96
MIM [14]	3.54±6.79	0.82±0.27	1.06±0.78	4.22±5.80	**2.23±0.83**	4.30±8.95	4.33±5.62	1.53±1.24	2.75±2.40
DINO [12]	4.20±6.20	0.85±0.35	1.31±1.19	3.96±5.92	3.50±2.44	**2.33±2.45**	4.41±5.05	1.43±0.94	2.75±2.14
VoCo [42]	2.66±3.73	0.93±0.45	1.23±1.25	4.03±5.84	3.52±3.09	3.47±5.62	3.59±4.14	1.24±0.75	2.58±1.78
Vox2Vec [13]	2.40±3.78	0.93±0.50	1.04±0.77	5.36±6.74	2.53±1.84	3.29±6.05	4.33±6.51	1.38±1.10	2.66±2.36
VolF (Ours)	**1.69±1.35**	**0.77±0.30**	**0.90±0.71**	**3.92±5.78**	2.69±1.30	3.38±5.79	**3.42±3.64**	**1.18±0.62**	**2.24±1.34**

training, but also largely improves the convergence speed. Specifically, VolF obtained an average validation Dice of 0.80 at 11k iterations, while training from scratch only converged to 0.77 at around 30k iterations.

For the kidney tumor in the KiTS dataset, compared with training from scratch, our VolF improved the average Dice from 70.72% to 79.31%, and reduced the average ASSD from 9.08 mm to 4.60 mm. In contrast, the best existing method Vox2vec [13] only obtained an average Dice of 77.72% with an ASSD of 5.99 mm.

Table III shows quantitative evaluation results on the Synapse dataset for abdominal organ segmentation from CT. The average Dice of training from scratch was 86.13%, and the corresponding value of the image restoration-based SSL methods and DINO [12] ranged from 86.79% to 87.51%, showing their limited performance improvement. In contrast, our method obtained an average Dice and ASSD of 88.67% and 2.24 mm respectively, and outperformed the existing SSL methods.

A visual comparison of different SSL methods on the three in-domain downstream datasets is shown in Fig. 6. It shows that training from scratch has challenges in obtaining accurate segmentation results of multiple organs and lesions when the training set is small, and our VolF leads to better results than existing SSL methods on these datasets. In addition, the validation Dice at different training iterations shown in Fig. 7(a) demonstrates that VolF led to much faster convergence than training from scratch, and is also better than Vox2vec [13] that is the best existing method in Table III.

2) Results on Out-Domain Downstream Datasets: We then show the results of different SSL methods on the SegTHOR and abdominal MRI datasets. Note that the two datasets involve unseen structure (chest OARs) or modality (MRI) during pretraining, respectively. Table IV shows quantitative evaluation results on the SegTHOR dataset. Training from scratch obtained an average Dice of 84.66%. Among the existing methods, Vox2vec [13] obtained the highest average Dice score of 86.48%. In contrast, our method improved it to 87.45% and outperformed the existing methods. Note that on esophagus that is the most difficult to segment, the improvement by our method is even larger. In addition, Fig. 7(b) shows that VolF has a faster convergence than Vox2vec and 'Scratch'. Specifically, VolF only needs 2k iterations to reach a validation Dice of 0.84, while Vox2vec and 'Scratch' require about 7k and 10k iterations to reach that performance, respectively.

Table V shows the results on the abdominal MRI segmentation task. It can be observed that learning from scratch obtained an average Dice of 80.03%. Patch Swap [44], Model Genesis [9] and VoCo [42] slightly improved it to 80.33%, 80.25% and 80.77%, respectively. Vox2vec [13], MIM [14] and DINO [12] had a better performance than them, with average Dice being 80.77%, 81.01% and 81.22%, respectively.

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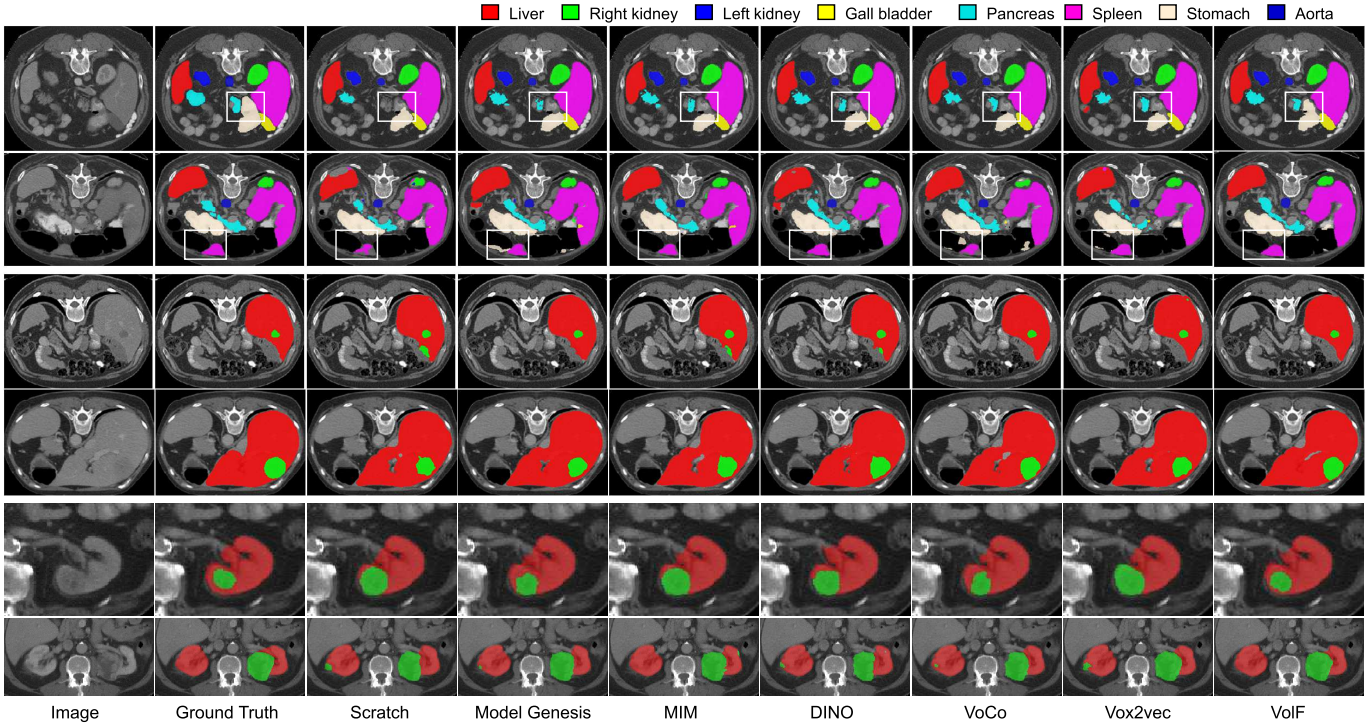

Fig. 6. Visual comparison between different SSL methods on in-domain downstream datasets. Rows 1-2 show results on the Synapse dataset, with white boxes highlighting the difference. Rows 3-4 and 5-6 show results on the LiTS and KiTS datasets, respectively, where green color represents the tumor.

TABLE IV

QUANTITATIVE COMPARISON BETWEEN DIFFERENT SSL METHODS ON THE SEGTHOR DATASET FOR CHEST ORGAN SEGMENTATION. THE BEST AND SECOND-BEST RESULTS IN EACH COLUMN ARE IN BOLD AND UNDERLINED, RESPECTIVELY

Method	Dice (%)					ASSD (mm)				
	Esophagus	Heart	Trachea	Aorta	Average	Esophagus	Heart	Trachea	Aorta	Average
Scratch	72.34±9.79	91.75±3.06	85.67±6.05	88.88±6.85	84.66±5.08	4.07±3.11	6.48±4.34	4.73±5.16	4.17±4.32	4.86±3.21
Patch Swapp [44]	72.99±10.22	93.28±2.29	87.31±6.09	88.36±8.42	85.49±5.70	2.83±2.24	3.62±3.47	3.03±4.58	4.89±6.44	3.59±3.70
Model Genesis [9]	74.13±10.04	92.37±3.56	86.79±7.80	89.17±8.08	85.62±5.71	2.75±1.48	4.42±4.71	2.13±2.82	3.20±4.21	3.13±2.56
MIM [14]	73.37±10.55	91.81±4.57	**87.97±5.49**	88.01±9.38	85.29±6.54	2.55±1.52	5.51±6.50	1.63±1.25	4.92±6.25	3.65±3.35
DINO [12]	73.24±11.11	92.64±3.35	87.46±7.02	90.46±5.30	85.95±5.55	2.79±1.68	4.94±6.91	2.29±2.98	2.50±2.79	3.13±3.03
VoCo [42]	72.58±10.50	93.21±2.61	87.78±5.46	90.17±4.64	85.94±4.77	3.60±3.49	4.08±3.89	1.63±1.62	2.71±2.49	3.01±2.57
Vox2vec [13]	74.25±9.76	93.09±2.47	87.21±6.20	91.35±4.04	86.48±4.58	2.52±1.84	4.21±3.20	**1.43±0.81**	2.28±3.43	2.61±2.06
VolF (Ours)	**76.80±9.01**	**93.79±1.47**	87.27±6.95	**91.93±3.40**	**87.45±4.17**	**2.36±1.65**	**3.48±2.80**	1.61±1.13	**1.85±1.33**	**2.31±1.27**

TABLE V

COMPARISON OF DIFFERENT SSL METHODS FOR ABDOMINAL ORGAN SEGMENTATION FROM MRI. IVC: INFERIOR VENA CAVAK, RAG: RIGHT ADRENAL GLAND, LAG: LEFT ADRENAL GLAND. THE LAST ROW REPORTS ASSD (MM), WHILE THE OTHER ROWS REPORT DICE (%). THE BEST AND SECOND-BEST RESULTS IN EACH ROW ARE IN BOLD AND UNDERLINED, RESPECTIVELY

Method	Scratch	Patch Swap [44]	Model Genesis [9]	MIM [14]	DINO [12]	VoCo [42]	Vox2vec [13]	VolF (Ours)
Liver	95.38±3.48	95.90±3.41	95.81±3.27	96.09±2.85	96.31±2.14	96.25±2.37	96.13±2.36	**96.35±2.10**
R Kidney	93.17±7.55	93.33±8.74	92.64±9.25	92.86±9.49	93.22±8.29	92.98±10.70	93.75±6.75	**94.06±6.70**
L Kidney	95.00±3.55	94.39±4.92	94.56±4.48	94.53±4.38	94.88±3.47	94.53±4.20	**95.07±3.79**	94.77±3.59
Spleen	88.80±18.62	89.73±18.29	89.65±18.67	90.57±17.14	**91.45±16.03**	89.92±18.21	89.73±18.07	91.37±16.59
Pancreas	80.72±13.83	80.50±13.11	80.49±13.73	81.68±12.55	**82.37±11.14**	81.00±11.71	81.51±13.66	82.15±11.61
Aorta	91.08±13.33	89.38±16.94	88.68±17.69	89.84±14.83	90.77±12.20	90.57±13.88	90.39±14.59	**92.26±7.94**
Stomach	80.52±14.22	78.82±16.86	80.15±14.25	80.76±13.99	82.60±12.91	81.30±15.14	80.60±16.22	**83.49±13.93**
IVC	78.15±24.00	77.28±24.67	76.89±25.10	79.16±22.14	79.19±21.99	79.08±22.49	77.92±23.48	**79.50±21.30**
RAG	59.95±21.03	60.94±23.02	59.25±22.15	60.83±21.63	60.06±20.80	60.00±23.52	62.59±19.86	**62.83±19.12**
LAG	67.03±22.68	67.89±18.31	**70.43±16.19**	69.79±17.24	70.34±17.92	69.54±19.04	70.43±20.38	70.09±17.69
Gallbladder	71.01±24.73	**79.98±20.79**	77.42±23.99	79.20±24.11	76.12±25.28	75.51±26.04	74.52±26.31	78.91±23.86
Esophagus	72.56±15.20	71.14±19.81	70.88±19.75	70.96±19.15	71.30±17.12	72.70±16.55	70.63±18.18	**73.30±16.44**
Duodenum	66.97±14.40	64.98±17.96	66.44±16.65	66.82±15.41	67.15±15.71	65.81±16.16	66.70±18.95	**69.11±14.22**
Avg Dice	80.03±8.81	80.33±9.00	80.25±9.50	81.01±8.72	81.22±8.07	80.71±8.57	80.77±9.18	**82.17±7.32**
Avg ASSD	3.00±4.55	2.61±2.36	2.64±2.29	2.27±1.81	2.22±1.68	2.26±1.58	2.63±2.87	**2.05±1.29**

In contrast, our method obtained an average Dice of 82.17%. In terms of average ASSD, compared with learning from scratch, our method reduced from it 3.00 mm to 2.05 mm, and also outperformed the existing SSL methods.

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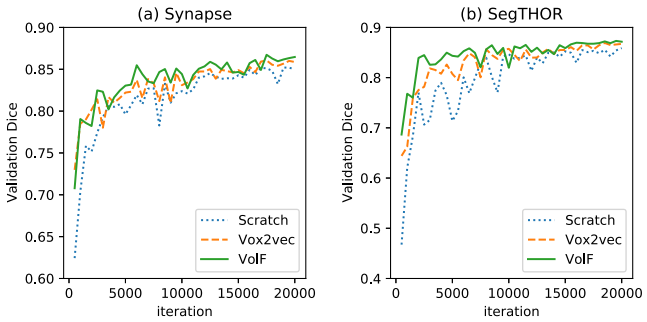

Fig. 7. Validation Dice during training obtained by scratch, Vox2vec and VolF on Synapse and SegTHOR datasets.

Fig. 8 shows a visual comparison between these methods. In the first two rows (SegTHOR dataset), 'Scratch' and the existing SSL methods obtained over-segmentation of the esophagus and aorta, respectively, while our method effectively avoids these mis-segmentations. In the last two rows that are for abdominal MRI segmentation, an under-segmentation of the liver and stomach can be observed in 'Scratch', respectively. VolF effectively improved the segmentation accuracy in these regions, and outperformed the other SSL methods.

D. Ablation Study

1) Hyper-Parameter Analysis: Our method has two key hyper-parameters: the number of foreground classes K and foreground ratio r for volume fusion. To investigate how they affect the performance, we first set K to 1, 2, 4, 8 and 16 respectively with $r = 0.7$, and then changed r in the range of 0.1 to 0.9 with $K = 4$. The SegTHOR and LiTS datasets were used as the downstream tasks. Fig. 9(a) shows the performance of VolF with different K values. Note that all the variants of VolF outperformed 'Scratch'. When K change from 1 to 4, the performance gradually increases, while $K = 8$ and $K = 16$ leads to some performance drop. This is mainly because that a too large K values makes the pretext task too difficult to learn, which limits the feature representation ability during pretraining. Fig. 9(b) shows the effect of r on the downstream segmentation performance, where the average Dice grows rapidly when r increases from 0.1 to 0.3, and continues to grow when r further increases from 0.3 to 0.7. Further increasing r to 0.8 and 0.9 obtains decreased Dice values, showing that both too low and too high r values result in sub-optimal performance. According to the best Dice in Fig. 9, we set $K = 4$ and $r = 0.7$ in the experiment.

2) The Role of Data Augmentation in VolF: We also analyzed how the data augmentation strategies of VolF affect the performance on out-domain downstream datasets. Our VolF with both intensity and spatial augmentations for pretraining was compared with three variants: 1) VolF (no-aug) means no data augmentation is used during pretraining; 2) VolF (i-aug) means only using intensity augmentation; and 3) VolF (s-aug) means only the spatial augmentation is used for pretraining. Results on the two out-domain downstream datasets are shown in Table VI. It demonstrates that both intensity and spatial augmentations in VolF are useful for dealing with out-domain downstream datasets. Note that VolF (i-aug) outperformed VolF (s-aug) on the AMOS-MRI dataset, showing the importance of intensity augmentation in VolF for dealing a different modality in the downstream dataset (CT vs MRI). In contrast, VolF (i-aug) is inferior to VolF (s-aug) on the SegTHOR dataset, demonstrating that spatial augmentation is more important for dealing with downstream datasets with a different body part (abdomen vs chest) in the same modality.

TABLE VI

EFFECT OF DATA AUGMENTATION STRATEGIES USED IN VOLF ON DEALING WITH OUT-DOMAIN DOWNSTREAM DATASETS. THE DICE AND ASSD ARE AVERAGED ACROSS ALL THE CLASSES IN EACH DATASET

Method	Dice (%)		ASSD (mm)	
	SegTHOR	AMOS-MRI	SegTHOR	AMOS-MRI
Scratch	84.66±5.08	80.03±8.81	4.86±3.21	3.00±4.55
VolF (no-aug)	86.85±5.13	80.50±8.95	2.73±1.60	2.80±3.22
VolF (i-aug)	87.02±4.23	81.61±8.60	2.66±1.49	2.35±1.86
VolF (s-aug)	87.13±4.10	81.07±8.92	2.45±1.47	2.61±2.47
VolF	**87.45±4.17**	**82.17±7.32**	**2.31±1.27**	**2.05±1.29**

E. Discussion

1) Improving Data-Efficiency in Downstream Datasets: In Table II to V, we have shown that VolF can effectively improve the segmentation performance on downstream datasets, especially for those with a small number of training samples, such as Synapse and SegTHOR datasets. To further show the data efficiency of our method in downstream tasks, we experimented with varying numbers of training samples for fine-tuning on the LiTS dataset: 10%, 20%, 50% and 100%, where 100% corresponds to 79 training samples. Fig. 10 shows the quantitative evaluation results of training from scratch and state-of-the-art SSL methods, including VolF, MIM [14], VoCo [42] and Vox2vec [13]. It can be observed that for liver tumor segmentation, VolF outperformed all the compared methods under different training sample ratios. What's more, with only 20% training samples for fine-tuning, VolF obtained an average Dice of 66.56%, which is higher than that of training from scratch with 100% training samples (65.42%). Compared with the later, VolF can obtain a lower average ASSD value with only 10% training samples (5.99 mm vs 6.23 mm). This shows the effectiveness of VolF in reducing the data requirement in downstream datasets compared with traditional supervised learning.

2) Benefit of Pretrained Decoder in VolF: An advantage of VolF is that its pseudo-segmentation-based pretext task makes features in both the encoder and the decoder be more transferable to downstream segmentation. In contrast, image restoration-based SSL methods [9], [14], [44] will make the decoder biased to image intensity prediction rather than semantic segmentation, and they usually only reuse the learned encoder for downstream segmentation tasks. To verify this, we compared VolF with Model Genesis [9] and MIM [14] under two different model initialization modes in downstream segmentation tasks: 1) 'Enc' mode that means only loading the pretrained encoder, and 2) 'Enc & Dec' mode that means loading both the pretrained encoder and decoder. As shown in Fig. 1, for Model Genesis, 'Enc & Dec' mode does not outperform 'Enc' mode during the downstream training process,

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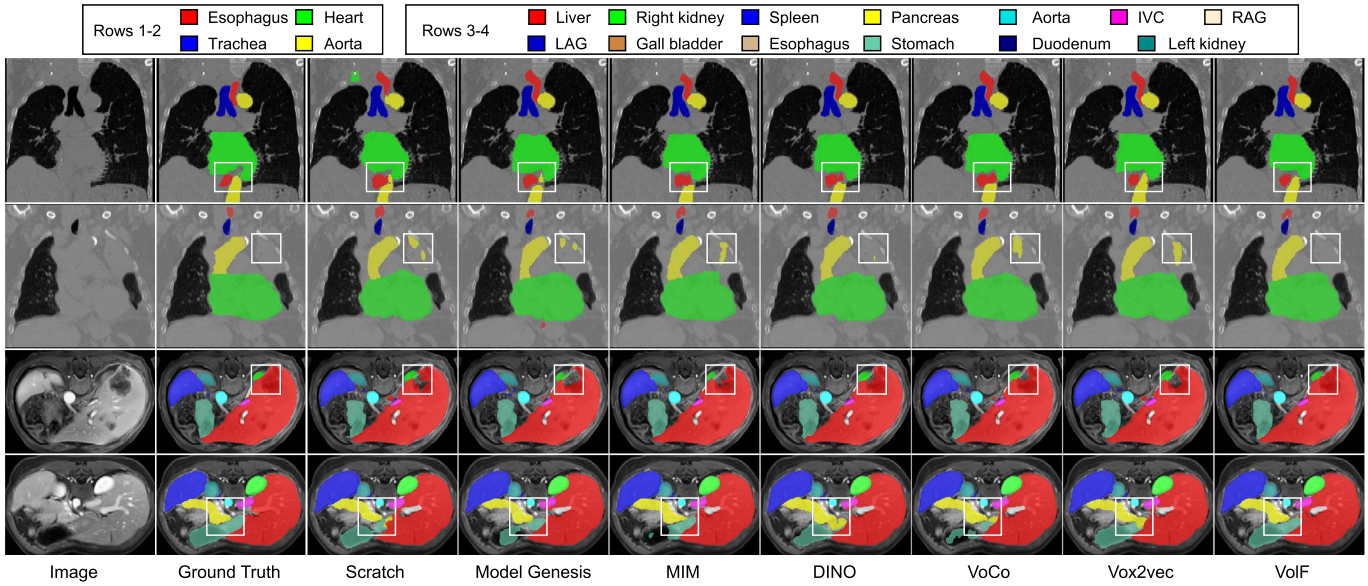

Fig. 8. Visual comparison between different SSL methods on out-domain downstream datasets. Rows 1-2 show results on the SegTHOR dataset, and rows 3-4 show results on the abdominal MRI dataset. White boxes highlight the local differences.

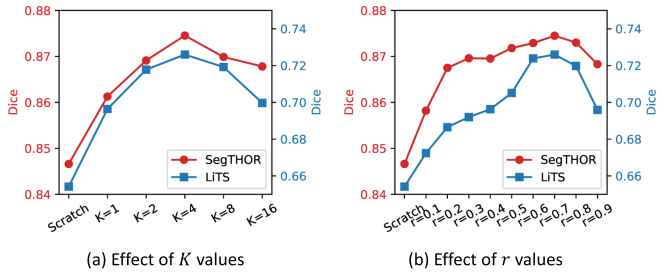

Fig. 9. Effect of hyper-parameters K and r on the average Dice on SegTHOR and LiTS datasets.

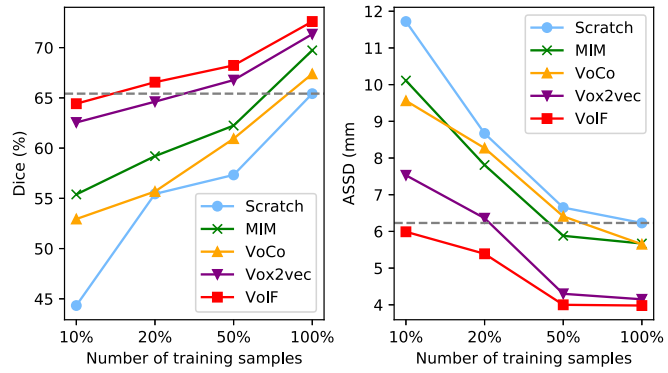

Fig. 10. Quantitative comparison of different SSL methods on the LiTS dataset with varying numbers of training samples for fine-tuning. The reported results are for liver tumor segmentation. The gray dashed lines show performance of training from scratch with 100% training samples.

and even leads to lower performance after convergence. In contrast, for VolF, 'Enc & Dec' mode outperforms 'Enc' mode obviously. In addition, the testing performance comparison on two different downstream datasets shown in Fig. 11 indicate that generally, 'Enc & Dec' mode does not outperform 'Enc'

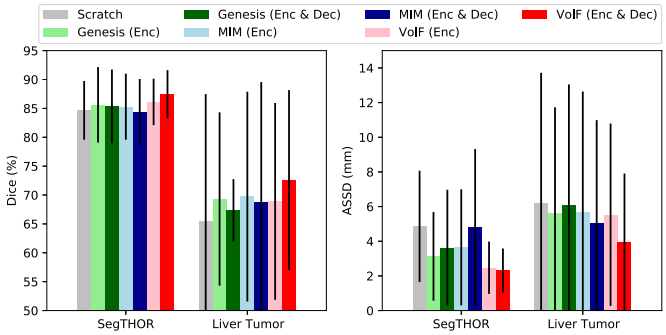

Fig. 11. Effect of pretrained encoder and decoder in Model Genesis, MIM and VolF on downstream segmentation datasets.

TABLE VII

EFFECT OF PRETRAINING WITH VOLF ON DIFFERENT NETWORK STRUCTURES. THE BEST AND SECOND-BEST RESULTS IN EACH ROW ARE IN BOLD AND UNDERLINED, RESPECTIVELY

Network	SSL	Dice (%)		ASSD (mm)	
		SegTHOR	Liver tumor	SegTHOR	Liver tumor
UNETR++	Scratch	83.36±6.34	64.81±20.94	4.28±2.47	6.27±6.17
	MIM [14]	85.67±4.26	65.96±21.35	3.31±1.92	5.94±5.67
	VoCo [42]	85.73±5.14	66.28±21.56	2.63±1.45	5.70±5.22
	VolF	**87.09±3.45**	**68.20±21.76**	**2.32±1.14**	**5.01±5.37**
nnFormer	Scratch	84.59±4.07	59.22±20.06	4.60±2.17	8.07±6.48
	MIM [14]	85.01±4.17	63.32±16.79	3.20±1.82	6.88±6.27
	VoCo [42]	85.37±4.43	62.13±16.70	3.47±2.05	6.57±6.11
	VolF	**86.39±2.63**	**66.96±17.99**	**2.40±1.26**	**5.14±4.88**

mode for Model Genesis and MIM, while 'Enc & Dec' mode is much better than 'Enc' mode for VolF, demonstrating that pretrained decoder of VolF is more useful than that of Model Genesis and MIM for downstream segmentation tasks.

3) Generalizability to Different Networks: Our VolF is a general self-supervised pretraining framework for medical image segmentation models, and does not rely on a specific network architecture. In addition to the FM-UNet that is a variant of 3D UNet [52], we also applied VolF to two

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Transformer-based networks: UNETR++ [24] and nnFormer [27], with a model size of 42.95 M and 149.22 M, respectively. The comparison between learning from scratch and pretraining with VolF for these networks on two downstream datasets are shown in Table VII. Note that as Vox2vec [13] was proposed for CNNs and not applicable to Transformers, we only used MIM [14] and VoCo [42] as the leading existing methods for comparison according to Table II and III. As shown in Table VII, on the SegTHOR dataset, VolF improves the average Dice by 3.73 and 1.80 percentage points for UNETR++ and nnFormer respectively. The performance gain is more obvious for liver tumor that is hard to segment, i.e., more than 3.4 percentage points for UNETR++ [24] and 7.5 percentage points for nnFormer [27] respectively in terms of average Dice. For both network structures, VolF outperformed MIM [14] and VoCo [42] that are state-of-the-art SSL methods based on image restoration and contrastive learning, respectively.

4) Pre-Training Time: As VolF pre-trains both the encoder and decoder with a segmentation loss, the pre-ptraining time is close to that of restoration-based methods. For example, for the FM-UNet-based backbone, the pre-training time for MIM [14], Model Genesis [9] and VolF was 13.34, 15.50 and 14.48 hours, respectively, which demonstrates the efficiency of our method for pre-training.

5) Limitation of this Work: This work also has some limitations that could be addressed in the future. First, as we used 1,112 CT scans for pretraining and the segmentation network is relatively small, it is of interest to apply VolF to pretraining datasets at a larger scale and larger models, which is expected to better unleash the power of VolF to build an SSL-based foundation model for medical image segmentation tasks [53], [54]. However, this will substantially increase the demand for computational resources. Second, we used a single modality for pretraining, and VolF could be extended to deal with multi-modal medical images. It is possible to fuse different modalities in VolF for the pseudo-segmentation task, and VolF can be combined with MAE [41] for enhancing modality-invariant feature learning. In addition, VolF is specifically designed for medical image segmentation tasks, and its effectiveness on other types of downstream tasks such as image classification and generation might be limited, which is of interest to investigate in future works.

V. Conclusion

We present a novel self-supervised learning framework based on Volume Fusion (VolF) for 3D medical image segmentation. The gap between pretraining and downstream segmentation is minimized by a pseudo-segmentation pretext task, where two sub-volumes are fused by a discretized block-wise fusion coefficient map, and the latter is converted into a segmentation label map so that standard fully supervised learning paradigm can be employed for pretraining. Experiments with an abdominal CT dataset for pretraining and five downstream segmentation datasets showed that VolF outperformed several state-of-the-art SSL methods with good generalizability to different network architectures, and the learned features are transferable to different body parts, lesions and modalities. It can achieve high performance and fast convergence with a small amount of data and annotations in downstream segmentation datasets. In the future, it is of interest to increase the diversity of modalities in the pretraining dataset and apply VolF to larger segmentation models.

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